

Eileen Dong

Seattle, WA

| eileendong1@gmail.com

[linkedin.com/in/eileen-dong](https://www.linkedin.com/in/eileen-dong) | github.com/eileendong | eileendong-one.vercel.app

Bachelor of Science in Computer Science, University of Washington · Expected June 2027

WORK EXPERIENCE

Software Engineer Intern, Axon

Seattle, WA | June 2026 – August 2026

- Built Kubernetes-based ephemeral preview environments for PAAS, giving every pull request an isolated live HTTPS gateway; served **41 PRs across 148 automated deployments in the Zero Touch Repository**.
- Designed a **controllerless GitOps architecture** using GitHub Actions, OCI Helm charts, and ArgoCD, where committing a single manifest provisions an environment and deleting it automatically tears down all associated resources.
- Diagnosed and eliminated cross-environment failures caused by globally shared Redis RBAC state by isolating each preview into its own **Kubernetes namespace, Temporal namespace, and Redis instance**.
- Secured execution of unreviewed PR code by removing application credentials, terminating TLS through an external Istio gateway, and validating **Kyverno policy enforcement and FIPS cryptography mode**.
- Built automated deployment safeguards including canary and end-to-end gates, idempotent teardown, PR-state reconciliation, and TTL-based orphan cleanup.
- Developed a **specification-first AI engineering workflow** using Claude skills and MCP servers across six task types, requiring claims to be grounded in source files, tickets, commits, or named owners; internal retrieval surfaced prior platform failures that directly informed the preview architecture.

Software Engineer Intern, H-E-B

Austin, TX | June 2025 – August 2025

- Replaced a legacy hardware alerting pipeline with a FastAPI microservice covering 13,000+ servers, reaching 99% delivery reliability and averting \$300,000+ in outage cost per incident.
- Migrated alert transport from SNMP over UDP to HTTP POST over TCP and added structured logging with Loguru and SQLite, persisting and querying 2,000+ alerts daily.
- Containerized the service with Docker and Kubernetes and instrumented it with Prometheus and OpenTelemetry, feeding real-time Grafana dashboards across the 13,000+ server fleet.

Student Information Technology Technician, University of Washington

Seattle, WA | January 2024 – Present

- Automated device imaging and inventory workflows across Windows, macOS, and Linux, cutting setup time by 40% while resolving 400+ hardware and software issues
- Deployed and maintained lab infrastructure; trained 5 technicians and standardized documentation across classroom and faculty systems.

Software Engineer Intern, then Research Intern, Southwest Research Institute

San Antonio, TX | 2022 – 2023

- Rewrote a command-line fluid thermodynamics simulator as a React application with CSV export, cutting engine test setup from minutes to seconds for 40+ engineers.
- Processed and digitized telemetry for the NASA Europa Clipper mission and supported Python sensor calibration testing across 2 summer terms.

PROJECTS

Automated Windows Workstation Deployment Infrastructure github.com/eileendong/bare-metal-provisioning

- Built an Ansible-driven bare-metal deployment system integrating with Microsoft Deployment Toolkit and PXE boot, eliminating 20+ manual steps and automating install of 20+ enterprise applications with Infrastructure as Code (Ansible, PowerShell, WinRM).

Roomies, Expense Splitting Application github.com/eileendong/roomies

- Built a React and Amazon Web Services (AWS) Amplify application using Cognito, Lambda, Textract, S3, and DynamoDB, parsing uploaded receipts into itemized splits with real-time sync over event-driven triggers.

Dub-Grind, AI Study Assistant github.com/eileendong/dub-grind2

- Built a Discord bot with LangChain, LlamaIndex, and the OpenAI API that summarizes lecture transcripts and generates adaptive quizzes from retrieval-indexed course material.

Committee Head, Senior Send-Off, Phi Sigma Rho engineering sorority. Directed a committee for a chapter event of 80+ attendees

EDUCATION

University of Washington, Bachelor of Science in Computer Science

Seattle, WA | Expected June 2027

Relevant coursework: Distributed Systems, Computer Security, Database Systems, Networks, Computer Architecture, Systems Programming, Algorithm Analysis, Linear Algebra, Natural Language Processing

TECHNICAL SKILLS

Programming Languages: Java, Go, Python, JavaScript, C, C++, SQL, Bash, YAML, HTML, CSS

Containers and Orchestration: Docker, Kubernetes, Helm, Istio, Envoy, Kyverno, External Secrets Operator, cert-manager, Role-Based Access Control (RBAC)

CI/CD, GitOps, and Infrastructure as Code: Continuous Integration and Continuous Delivery (CI/CD), GitOps, ArgoCD, GitHub Actions, GitHub Apps, GitLab CI/CD, Terraform, Ansible, PowerShell, Git, Linux

Cloud Platforms: Microsoft Azure (AKS, ACR, Key Vault, Workload Identity), Amazon Web Services (AWS): Lambda, S3, DynamoDB, Cognito, IAM, Amplify, Textract

Backend and Distributed Systems: FastAPI, REST APIs, Node.js, React, Pydantic, Temporal, Redis, SQLite, microservices, OAuth 2.0, JSON Web Tokens (JWT), mutual TLS

Observability and AI: Prometheus, Grafana, OpenTelemetry, Claude, Model Context Protocol (MCP), agent skills, LangChain, LlamaIndex, OpenAI